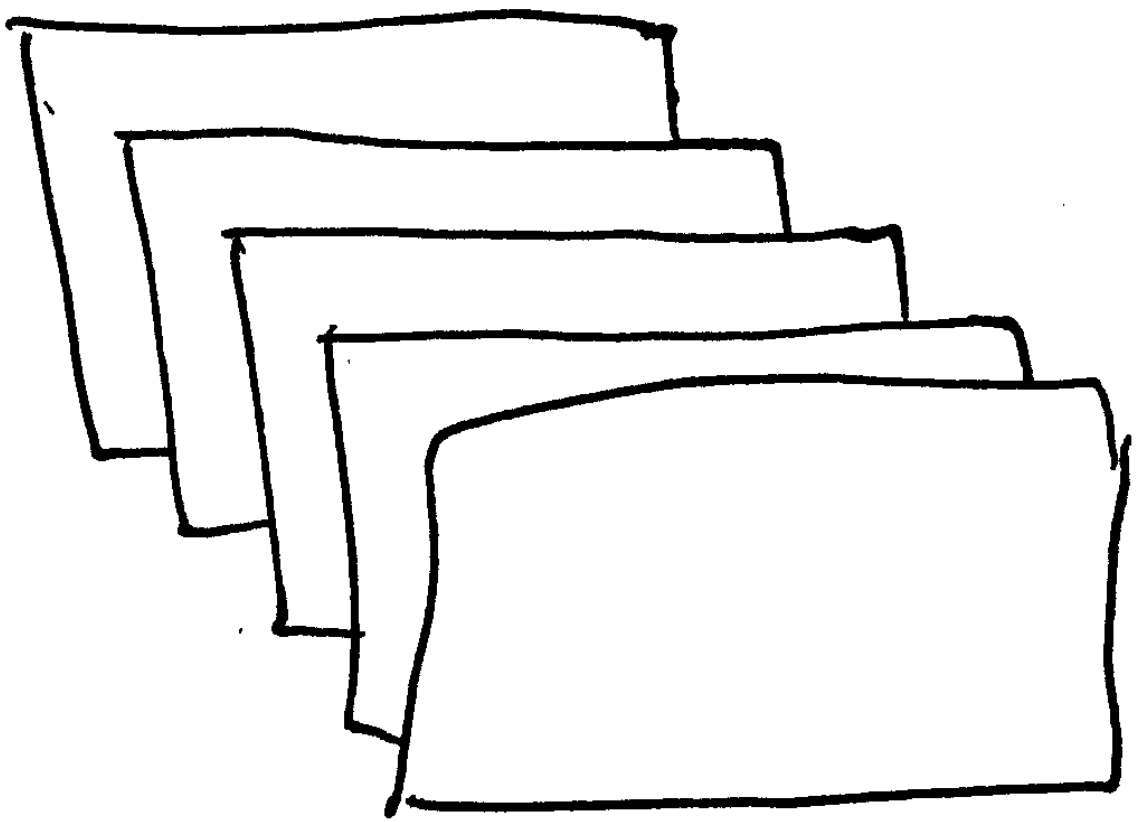
Analysing video sequences

2026-03-20

SDU

What is a video

A sequence of frames



Limited change from one frame to the next I(x, y, t)

Things to know about a video

- Object tracking
- Track camera motion
- Background subtraction
- Shot boundary detection
- Motion segmentation

Suggested code structure

Frame viewer

```
import numpy as np
import cv2
filename = 'input/remember.mkv'
self.cap = cv2.VideoCapture(filename)
while(1):
    ret, frame = self.cap.read()
    if ret is not True:
        break
    # Process frame
    cv2.imshow('frame',frame)
    # Deal with key presses
    k = cv2.waitKey(30) & 0xff
```

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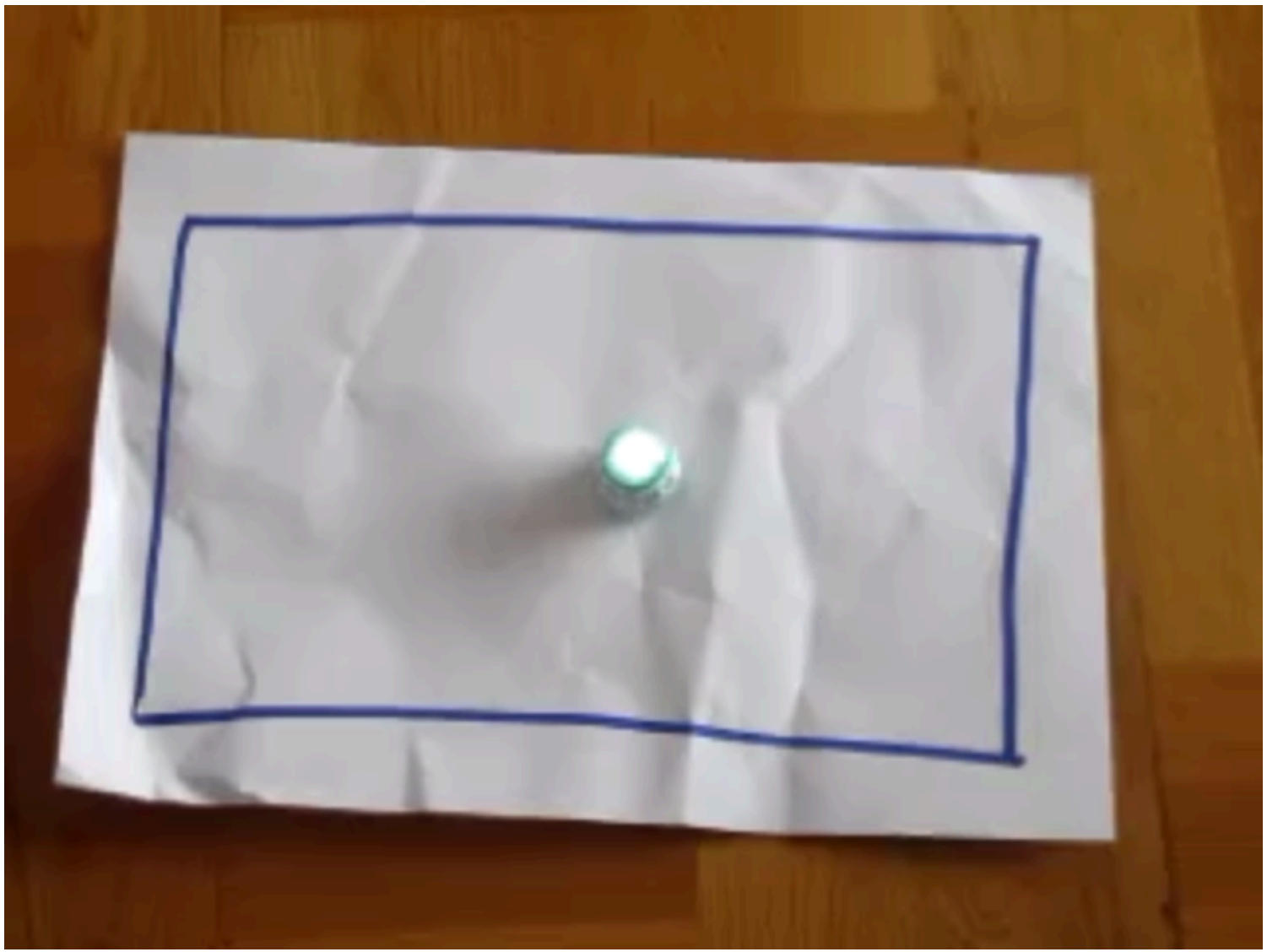
Frame viewer refactored

```
import numpy as np
import cv2
class FrameIterator():
    def __init__(self, filename):
        self.cap = cv2.VideoCapture(filename)
    def frame_generator(self):
        # Define a generator that yields frames from the video
        while(1):
            ret, frame = self.cap.read()
            if ret is not True:
                break
            yield frame
        self.cap.release()
    def main(self):
```

Color based tracking

Color based tracking



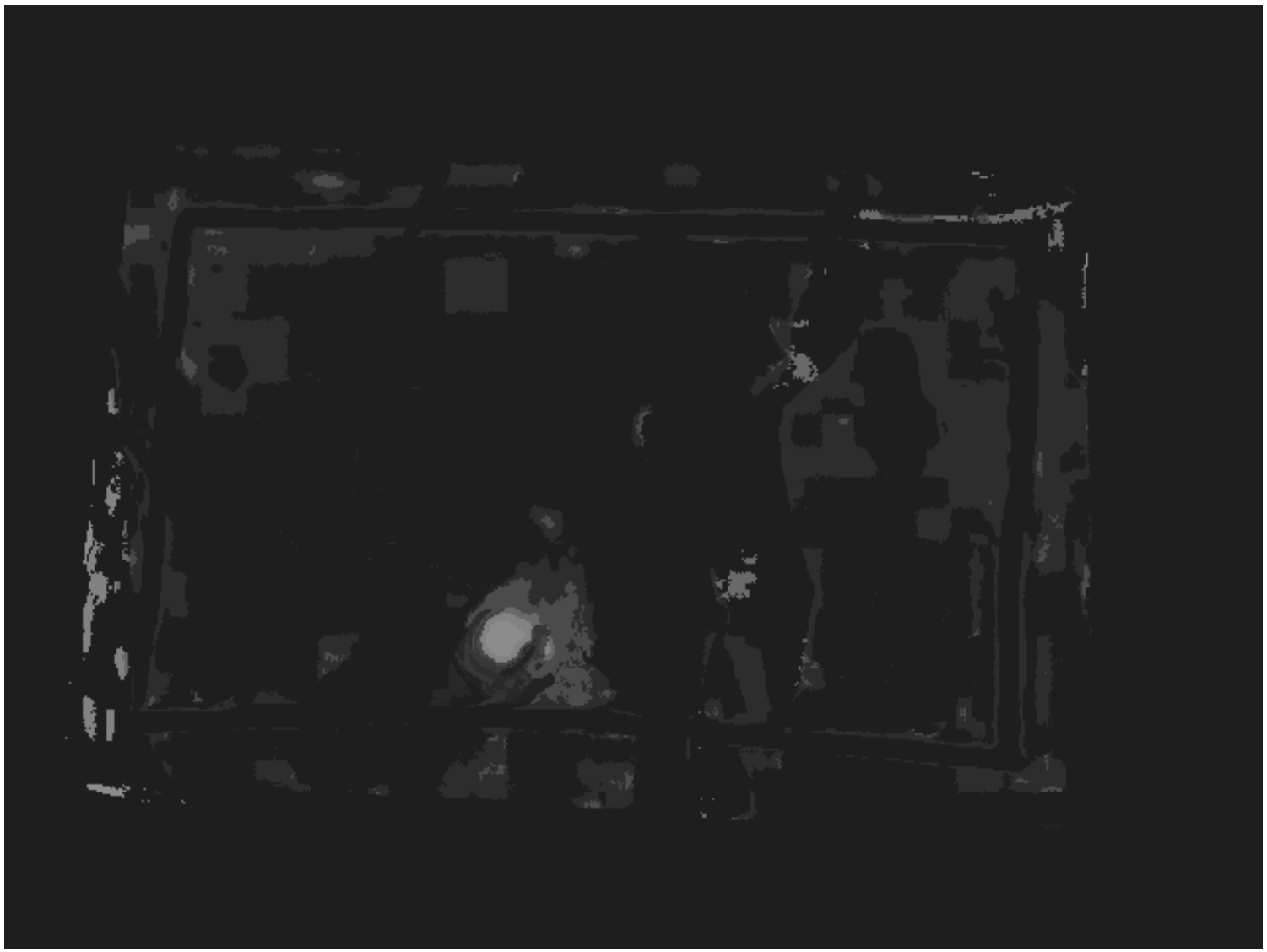
Preprocessing of image

Histogram backprojection

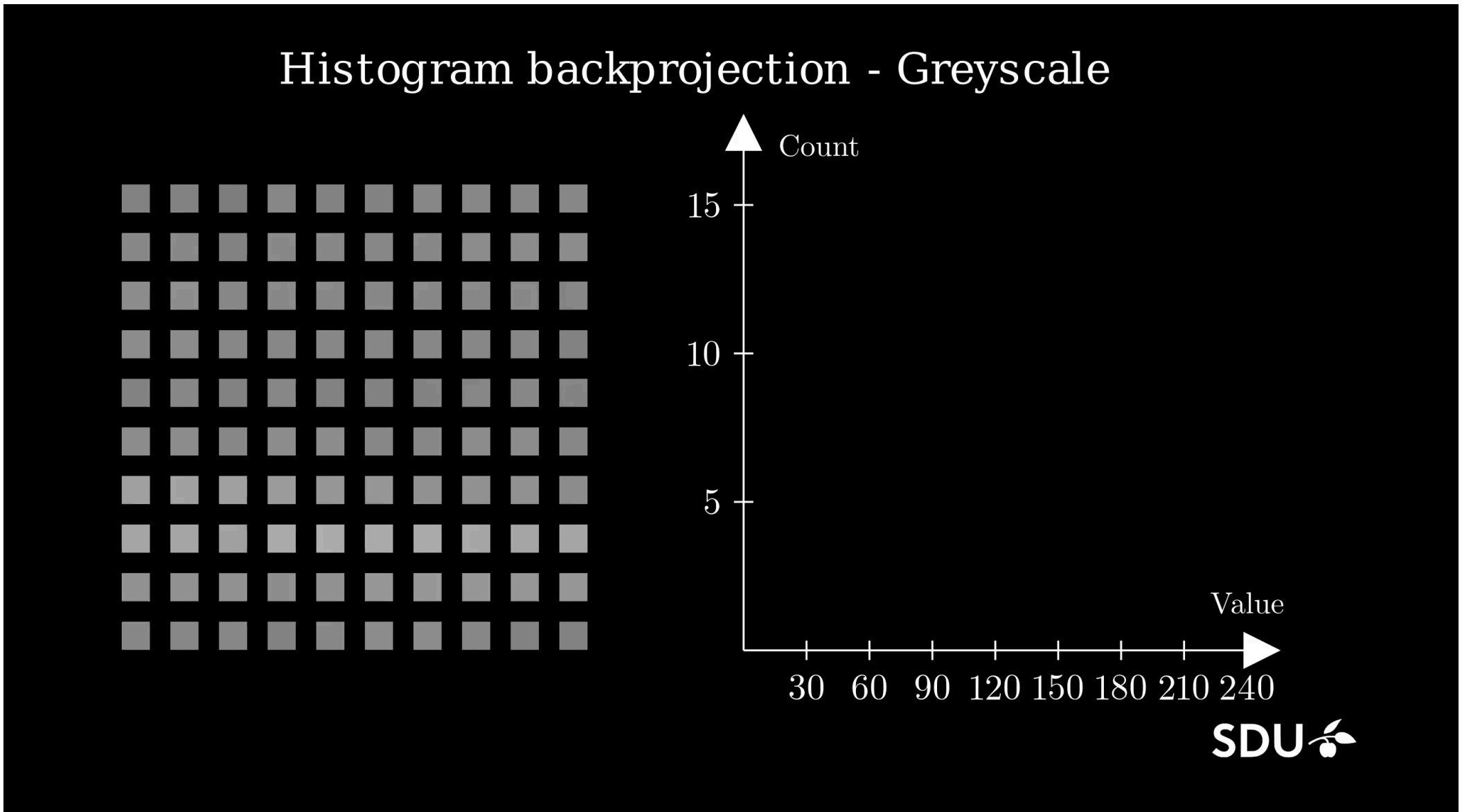
- Given a known reference object / set of pixels
- Locate pixels in the new image with similar colors as the reference object

Histogram backprojection

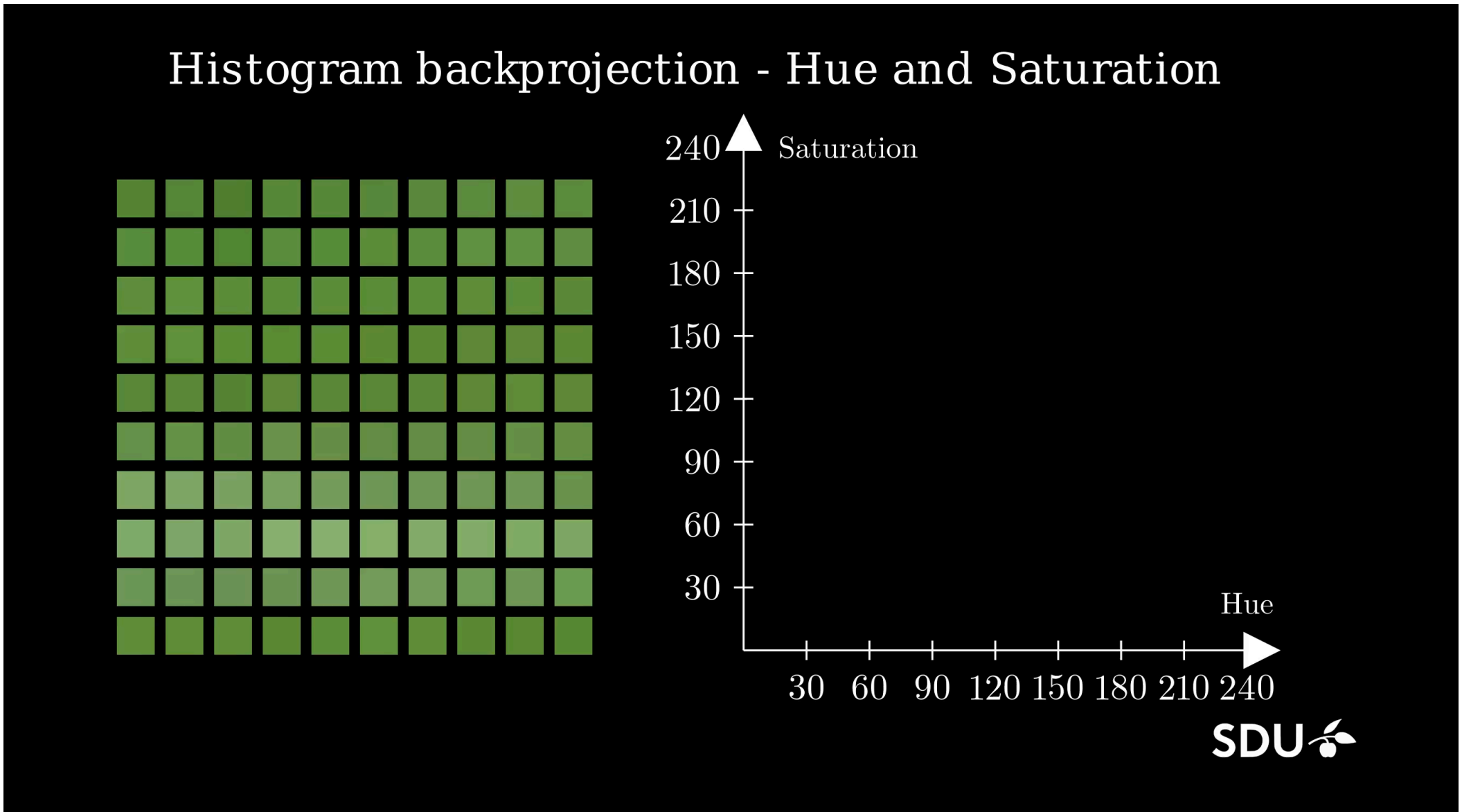
Histogram backprojection



Animation - grayscale



Animation - hue and saturation



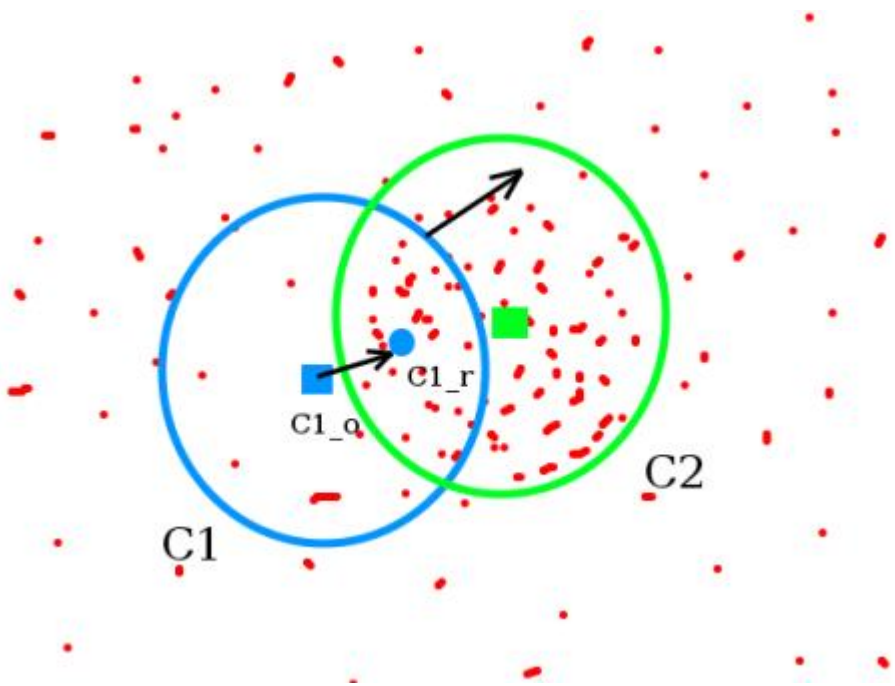
Tracked position

Online example

http://henrikmidtiby.github.io/downloads/objecttracker/objecttra

Example video: 2019-02-15 15.42.26.mov

Meanshift

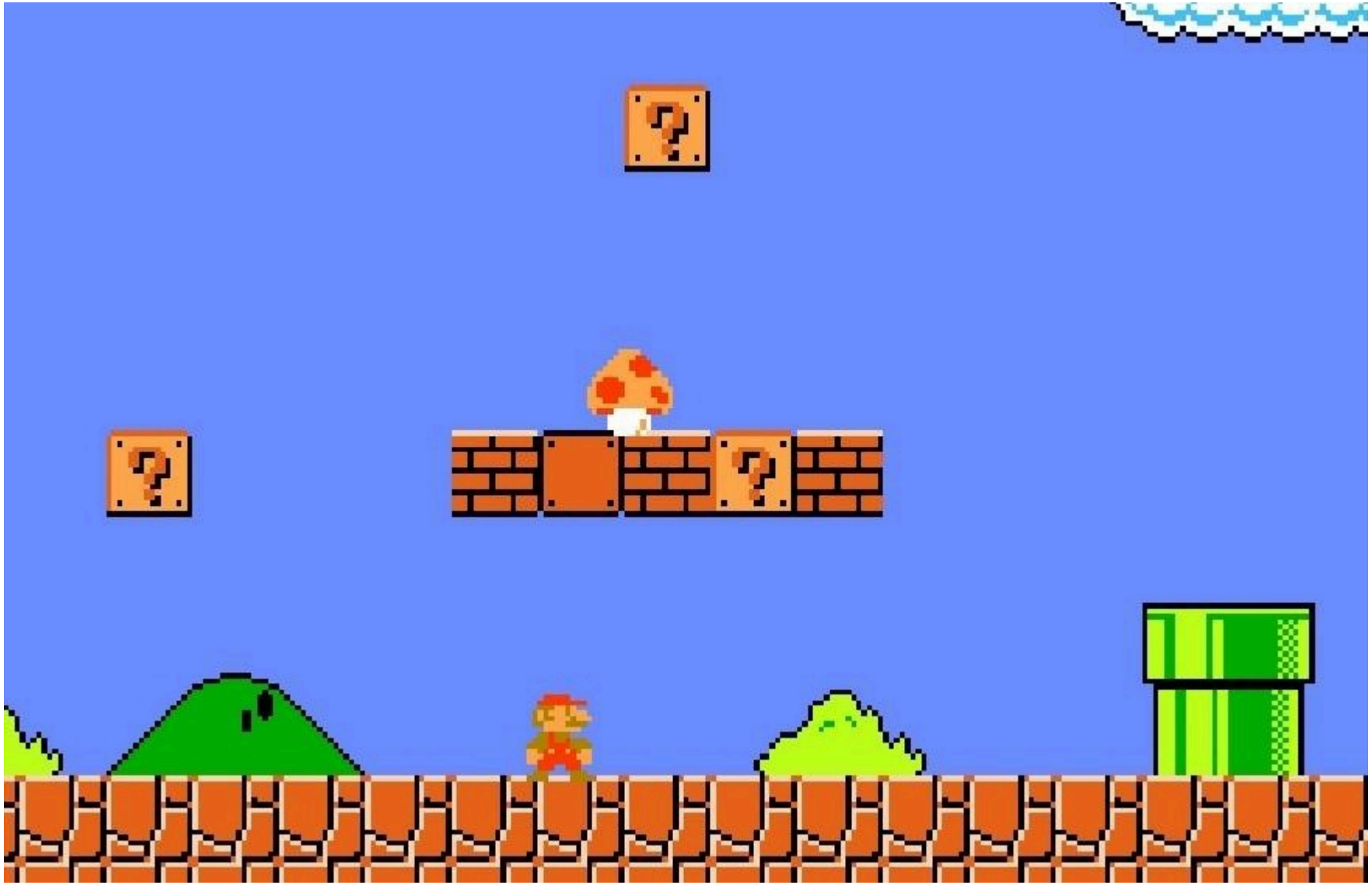


Camshift

Camshift is a modification of meanshift, that also considers the size of the tracked object.

Template matching

Locate a small image (a template) in a large image



SDU

Cross correlation

Slide the template over the image, for each position, calculate a match score $R(x, y)$.

$$R(x, y) = \sum_{x', y'} \left(T(x', y') - I(x + x', y + y')\right)^2$$

- T template
- I image
- x coordinate (global)
- y coordinate (global)
- x' coordinate (local)
- y' coordinate (local)

Size of template and image

Image: 566 x 877 x 3

Template: 48 x 48 x 3

Match: 519 x 830

566 - 48 + 1 = 519

877 - 48 + 1 = 830

Using cross correlation to locate a template in an image

```
# Applying template matching
result = cv2.matchTemplate(image, template, cv2.TM_SQDIFF)
# Locating the best match
minVal, maxVal, minLoc, maxLoc = cv2.minMaxLoc(result, None)
```

Result of template matching

When does template matching work?

- Template matches target
- Same size
- Same orientation

Motion estimation

Motion estimation techniques

Feature based methods

- corners
- textured areas

Dense methods - Assume that object brightness is constant - only good for small movements

Dense - Optical flow

Is the apparent motion of objects / points

Contraints

Brightness constancy contraints

$$I(x, y, t) = I(x + u, y + v, t + 1)$$

Small motion

$$I(x + u, y + v) = I(x, y) + \frac{dI}{dx} \cdot u + \frac{dI}{dy} \cdot v + \text{higher order}$$

Calculations

For t or $t+1$

$$I_x = \frac{dI}{dx}$$

$$\begin{aligned}
0 &= I(x+u, y+v, t+1) - I(x,y,t) \\
&\sim I(x,y,t+1) + I_x \cdot u + I_y \cdot v - I(x,y,t) \\
&\sim [I(x,y,t+1) - I(x,y,t)] + I_x \cdot u + I_y \cdot v \\
&\sim I_t + I_x \cdot u + I_y \cdot v \\
&\sim I_t + \nabla I \cdot < u, v >
\end{aligned}$$

Problem with more unknowns than equations

For each pixel there are

- two unknowns
- one constraint equation

Where to find additional constraints?

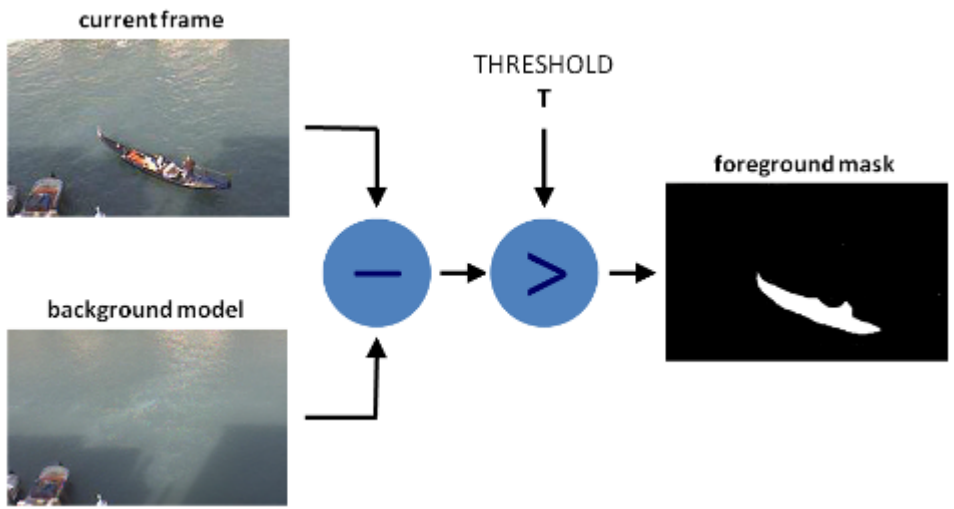
Solving the aperture problem

Track eg. a 5x5 region and assume that u and v are the same for all pixels in this region.

B. D. Lucas and T. Kanade (1981), An iterative image registration technique with an application to stereo vision. Proceedings of Imaging Understanding Workshop, pages 121–

Background estimation / subtraction

Background subtraction



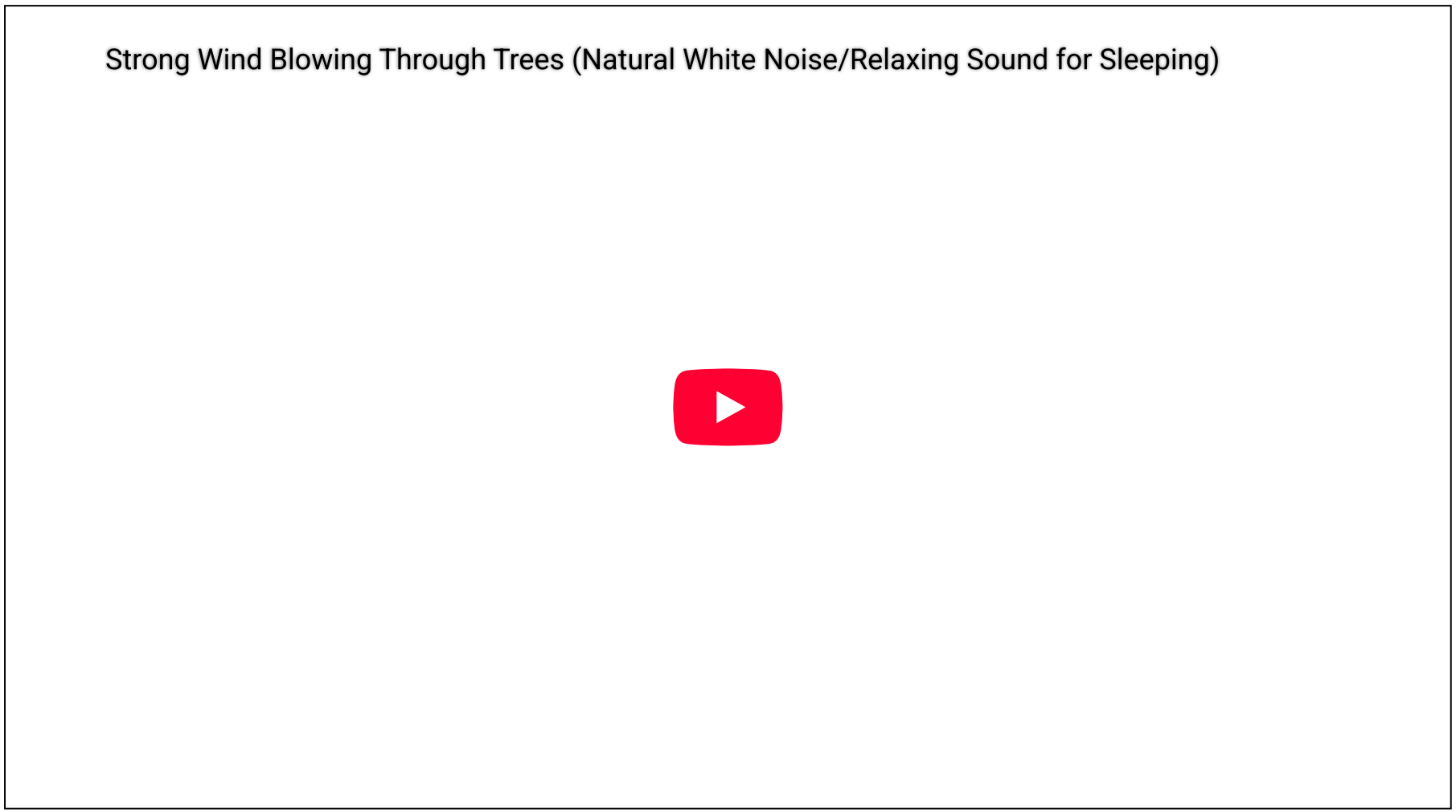
Simplest possible ?!

Remember the previous frame and use that as a model of the background

Leads to a very rough background model

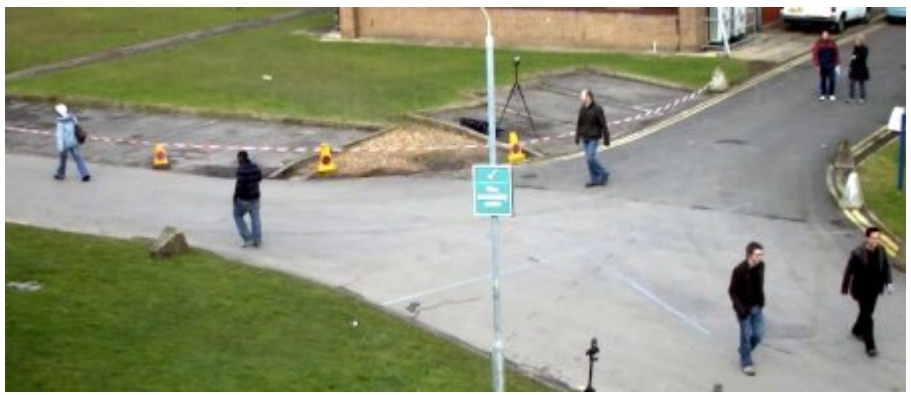
Not suitable when objects are moving slowly

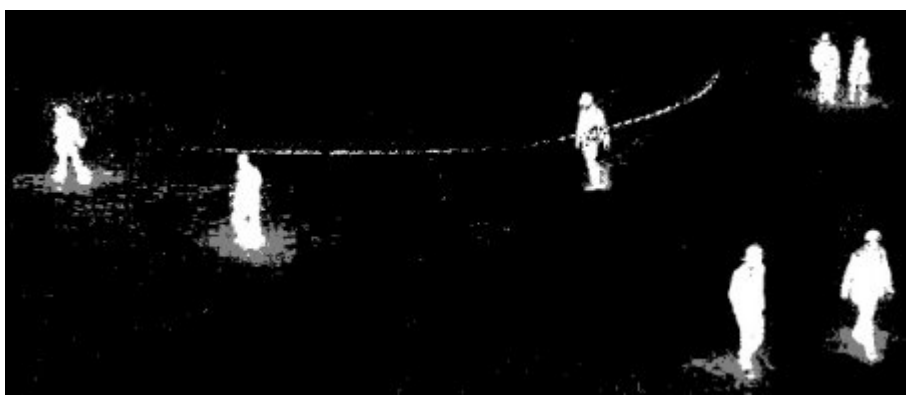
Background estimation



Model how each pixel change value over time

Examples - BackgroundSubtractorMOG2





Motion detection miniproject

Input video

2017_06_22_1301 Roersvej støder op til Vesterbro

Outro

Course overview

2026-02-01

- Python and 2D image processing
- Camera in space
- Optimization
- Fiducial markers

2026-03-01

- Camera calibration mini project
- Feature detectors and video stabilization
- Motion detection mini project
- Visual odometry

2026-04-01

- Visual odometry mini project
- Orthomosaics
- Pumpkin counting mini project

2026-05-01

- Bag of Visual Words / AI overview

Quote

“Of course, it is very important to be sober when you take an exam. Many worthwhile careers in the street-cleansing, fruit-picking and subway-guitar-playing industries have been founded on a lack of understanding of this simple fact.”

— Terry Pratchett, Moving Pictures